

Three types of IHRCs in Korean and their interpretations

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1. Introduction: IHRCs vs. EHRCS

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1 Introduction: IHRCs vs. EHRCS

- (1) inho-nun [*mina/uyca-ka* nemeci-nun]_{CP} **kes**/**yeca-ul*]_{NP} cap-ass-ta.
Inho-top Mina/chair-nom fall-*adn* thing/woman-acc hold-*pst-dec*
(IHRC)
'Inho held {Mina/a chair} when she/it was falling down.'
- (2) a. inho-nun [*e* nemeci-nun]_{CP} **yeca-lul**]_{NP} cap-ass-ta. (EHRC)
Inho-top fall-*adn* woman-acc hold-*pst-dec*
'Inho held a woman who was falling down.'
- b. inho-nun [*e* nemeci-nun]_{CP} **kes-ul**]_{NP} cap-ass-ta. (EHRC)
Inho-top fall-*adn* thing-acc hold-*pst-dec*
'Inho held {a thing} /*{a person} that was falling down.'

Property 1: no gaps in IHRCs (definitional)

Property 2: NPs modified by IHRCs are always headed by *kes*.

Property 3: *kes* denotes people/things with IHRCs; it denotes things with EHRCS.

abbreviations: acc(usative case), adn(ominal), cad(sative), c(o)mp(lementizer), dec(larative), hor(tative), imp(erative), imp(er)fective aspect, int(errogative), mod(al), n(or)m(ina)l(izer), nom(inative case), pass(ive), p(ost), p(er)fect, p(ural), pr(o)g(ressive), relativizer, r(e)tr(ospective), top(ic).

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Temporal restriction

- (3) inho-nun [*lecey e* yeki-ey iss-te-n]_S sakwa-lul]_{NP} onul mek-ess-ta.
Inho-top yesterday here-at be-*rtt-adn* apple-acc today eat-*pst-dec*
(EHRC)
'Today Inho ate the apple which had been here yesterday.'
- (4) a. *inho-nun [*lecey yeki-ey* sakwa-ka iss-te-n]_S kes-ul]_{NP} onul
Inho-top yesterday here-at apple-nom be-*rtt-adn* thing-acc today
eat-*pst-dec* (IHRC)
'Today Inho ate the apple which was here yesterday.'
- b. inho-nun [*yeki-ey* sakwa-ka iss-nun]_S kes-ul]_{NP} mek-ess-ta. (IHRC)
Inho-top here-at apple-nom be-*adn* thing-acc eat-*pst-dec*
'Inho ate the apple which was here.'

Property 4: Events in IHRCs must be simultaneous with those in the embedding clauses.

Jae-Il Yeom

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1. Introduction: IHRCs vs. EHRCS

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Problems with previous observations

Problem 1: IHs of IHRCs in subject positions denoting only things

- (5) a. mina-nun [*gam-i* tteleci-n]_S kes-ul cal ik-hi-ess-ta.
Mina-top persimmon-nom fall-*adn* thing-acc well ripe-*can-pst-dec*
'Mina ripened well persimmons which fell on the ground.'
- b. [*gam-i* tteleci-n]_S kes-i cal ik-ess-ta.
persimmon-nom fall-*adn* thing-nom well ripe-*pst-dec*
'Persimmons which fell on the ground were more ripe.'
- (6) a. inho-nun [*totwuk-i* tali-lul kenne-nun]_S kes-ul
Inho-top thief-nom bridge-acc cross-*adn* thing-acc
pwtuthcap-ass-ta.
catch-*pst-dec*
'Inho caught a thief who was crossing the bridge.'
- b. ?[*totwuk-i* tali-lul kenne-nun]_S kes-i pwtuthcap-hi-ess-ta.
thief-nom bridge-acc cross-*adn* thing-nom catch-*pass-pst-dec*
'A thief was caught who was crossing the bridge.'

In a subject position, the IH of an IHRC must be a thing.

Problem 2: Cases where the relevancy condition is not satisfied

- (7) [kuttay cha-ka eyncin-i kkeci-iss-nun] kes-ul si-long-ul
then car-*nom* engine-*nom* stop-*impr-adv* thing-*acc* starter-*acc*
kel-**{ess/?n.kes.i}**-ta.
start-**{pst/mod}**-*dec*
'(U) {started/will start} the engine of the car whose engine was turned off then.'
- (8) [cikum cha-ka eyncin-i kkeci-iss-nun] kes-i **nayil**
now car-*nom* engine-*nom* stop-*impr-adv* thing-*nom* tomorrow
chwulcenha-l cha-i-ta.
participate-*adv.mod* car-be-*dec*
'The car whose engine is now turned off is going to participate in the race **tomorrow**.'
- (9) inho-nun [mina-ka cikum ku kamca-lul thwuyki-nun]^{CP} kes-ul^{NP} **nayil**
Inho-*top* Mina-*nom* now the potato-*acc* fry-*adv* thing-*acc* tomorrow
mek-ul,kesi-ta.
eat-*mod-dec*
'Inho will eat **tomorrow** the potato which Mina is frying **now**.'
- (10) [kkun-i sinchwukseng-i-iss-nun] kes-un il **nyen hwu-ey**-to an
rope-*nom* elasticity-*nom* have-*adv* thing-*top* one year later-at-also not
kkuneci-l,kesi-ta.
be.cut-*mod-dec*
'The / A rope which is elastic will not break even in a year.'

Pseudo IHRCs¹

- (11) inho-nun [mina-ka toktokha-n3] kes-ul **chingchanhay**-ss-ta.
Inho-*top* Mina-*nom* smart-*adv* thing-*acc* praise-*pst-dec*.
'Inho praised Mina's smartness.'
- (12) inho-nun [mina-ka pwumno-lul ilh-un] kes-ul **wuylohay**-ss-ta.
Inho-*top* Mina-*nom* parents-*acc* lose-*adv* thing-*acc* console-*pst-dec*
'Inho consoled Mina when she lost her parents.'

The verbs in the IHRCs can take events as their arguments:

- (13) a. Inho-nun [senhayng-ul ha-n2] **ai-lul chingchanha**(y)-ss-ta.
Inho-*top* good.deed-*acc* do-*pst* kid-*acc* praise-*pst-dec*
'Inho praised the **kid** who did a good deed.'
- b. Inho-nun ai-ka ha-n2 **senhayng-ul chingchanha**(y)-ss-ta.
Inho-*top* kid-*nom* do-*pst* good.deed-*acc* praise-*pst-dec*
'Inho praised the kid's **good deed**.'
- (14) a. inho-nun pwumno-lul ilh-un] **mina-lul wuylohay**-ss-ta.
Inho-*top* parents-*acc* -lose-*adv* Mina-*acc* console-*pst-dec*
'Inho consoled **Mina**, who lost her parents.'
- b. inho-nun mina pwumno-uy **cwukum-ul wuylohay**-ss-ta.
Inho-*top* Mina parents-of death-*acc* console-*pst-dec*
'Inho consoled the **death** of Mina's parents.'

¹M.J. Kim (2004) takes these to be IHRCs, but they are not, based on the discussion that follows.

Organization of the presentation

- The structure of an adnominal clause and the meanings of the morphemes involved
 - internal structure of an IHRC and the meanings of adnominal endings
 - general properties of the head noun *kes*
- Three types of IHRCs and their characteristics
 - necessity of the distinction of stages and individuals
 - existential disclosure for adnominal clauses in general
- Meaning of *kes*
- Distributions of three types of IHRCs
- No necessity of the relevancy condition
- Interpretation of strong quantifiers as IHs of IHRCs
- Summary

2 IHRCs as adnominal clauses in general

2.1 Internal structures of adnominal endings

(15)

Aspect	Tense	Modal	Retro	C _{rel}	realizations
$\Pi_{impr} / \emptyset_{Asp}$	$\emptyset_{T_i}$	$\emptyset_{M_i}$	$\emptyset$	(u)n	nun, (u)n3
$\emptyset_{prf}$	$\emptyset_{T_i}$	$\emptyset_{M_i}$	$\emptyset$	(u)n	(u)n1
$\emptyset_{asp}$	$\emptyset_{pst_i}$	$\emptyset_{M_i}$	$\emptyset$	(u)n	(u)n2
$ess / \emptyset_{Asp}$	$ess_{T_i} / \emptyset_{T_i}$	(u) _i	$\emptyset$	$\emptyset$	((e)ss(-ess)-(u))
$ess / \emptyset_{Asp}$	$ess_{T_i} / \emptyset_{T_i}$	keys _s / $\emptyset_{M_i}$	te	(u)n	((e)ss(-ess)-te-n, keys _s -te-n

(relative) present imperfect: *nun*(non-stative verbs), *(u)n3*(stative verbs)
(relative) present perfect: *(u)n1*(only with non-stative verbs)
(relative) past: *(u)n2* (non-stative verbs)
modality: *-(u)l, keys_s*(expressing futurity)

- (16) [[chayk-i suywu-**{n3 / (u)}** / **ess-ul** / **ess-te-n**] kes-ul] kol-ala.
book-*nom* easy-be-*{adv / adv.mod / pst-adv.mod / pst-*th-re*}* thing-*acc* choose-*impr*
'Choose books which {are, will be, must have been, was} easy.'

Our discussions are centered around the adnominal endings *nun*, *(u)n1/2* and *(u)n3*.

Meanings of the morphemes in adnominal endings

An aspect changes a property of events into a property of times.

- (17) $[(\langle n \rangle u) n_{mnp}]^{w,g} = \lambda P \lambda t \lambda w \exists e [t \subseteq \tau(e) \wedge P(e)(w)]$, where e is an event variable.
 $\tau(x)$ is the time during which x exists or lasts.)

$[\emptyset_{Asp}]^{w,g} = (i) \lambda P \lambda t \lambda w \exists e [\tau(e) \subseteq t \wedge P(e)(w)]$, where e is an event.

(ii) $\lambda P \lambda t \lambda w \exists e [t \subseteq \tau(e) \wedge P(e)(w)]$, where e is a state.

$[\emptyset_{prf}/\text{ess}_{prf}] = \lambda P \lambda t \lambda w \exists Q \exists e' [\exists e [\tau(e) \preceq t \wedge P(e)(w)]] \wedge \tau(e') = (t) \wedge Q(e')(w)]$
 $(P$ is a previous event and Q is a resulting state.)

- i. $t = (t':t'')$ iff t begins with t' and ends with t'' , if t' (or t'') is missing, t begins (or ends) at an arbitrary time.
- ii. $\exists e [P(e)$ entails $\exists e' [Q(e')]$.

Tense and mood are analyzed like pronouns whose values are given by an index or an assignment.

- (18) $[\emptyset_T]^{w,g} = t_r$ if $t_r = g(i)$; undefined otherwise.

$[\emptyset_{pse}/\text{ess}_{pse}]^{w,g} = t_r$ if $t_r = g(i) \wedge t_r \prec t_w$; undefined otherwise.

$[\emptyset_{Mw}]^{w,g} = w'$ (w : a variable, w' : a particular world)

2.2 Properties of *kes*

1. It is defective in that it *must* be preceded by an adnominal expression.

- (19) **na-nun kes-tul-ul sa-ssta.*
 I-top thing-pl-acc buy-psf-dec
 'I bought things.'

2. It always denotes things, not people in other constructions.

- (20) *inho-nun [na-lul salangha-nun] {salam/?kes}-kwa kyelhonha-ko*
 inho-top I-acc love-adv person/thing-with marry-and
 sipb-ta.
 like-dec
 'Inho wants to marry a person/thing that loves me.'

Jhang (1994) observed that NPs modified by IHRCs cannot be used as indirect objects.

- (21) ?*mina-nun [[haksayng-i chacao-ni] kes]_{NP}-eykey cilmwun-ul*
 Mina-top student-nom come-adv thing-to question-acc
 ha(y)-ss-ta.
 do-psf-dec
 'Mina asked a question of the student who came.'

3. It can denote various things provided by embedded clauses which do not correspond to gaps.

- (22) a. *na-nun [[mina-ka ka-nun] kes-ul]_{NP} al-ass-ta.* (fact)
 I-top mina-nom go-adv thing-acc know-psf-dec
 I knew that Mina was going.
 b. *na-nun [[mina-ka ka-nun] kes-ul]_{NP} po-ass-ta.* (event)
 I-top mina-nom go-adv thing-acc see-psf-dec
 I saw Mina going.
 c. *na-nun [[mina-ka ka-nun] kes-ul]_{NP} cap-ass-ta.* (object)
 I-top mina-nom go-adv thing-acc catch-psf-dec
 I caught Mina when she was trying to go.

Here adnominal clauses have no gaps which correspond to the denotations of the NPs with *kes*.

Assumption: The meaning of *kes* is the same all the time.

Implications:

- NPs modified by IHRCs do not denote people.
- An IHRC is not a special construction, but is interpreted in the same way as cases where adnominal clauses denote properties of facts or events.

3 Three types of IHRCs

3.1 An IHRC of Type I as denoting a property of stages

- (23) *inho-nun [mina-lul]_{NP} mit/mil-ess-ta.*
 Inho-top Mina-acc believe/push-psf-dec
 'Inho liked/caught Mina.'
 (24) *inho-nul [[mina-ka chwumchwu-nun] kes-ul]_{NP} ?mit/mil-ess-ta.*
 Inho-top Mina-nom dance-adv thing-acc believe/push-psf-dec
 'Inho loved/caught Mina when she was dancing.'

The verb *mit* 'believe' requires a person as the object, but the NP with the IHRC 'Mina at the time she was dancing' is not a person.

- (25) *na-nun [ku kwusul-i kwulleka-nun] kes-ul cap-ass-ci [ku*
 I-top the marble-nom roll-adv thing-acc catch-psf-nml the
kwusul-i s(u)-e-iss-nun] kes-ul cap-ci anh-ass-ta.
 marble-nom stop-imp-adv thing-acc catch-nml not-psf-dec
 'I caught the marble when it was rolling, not the marble when it was standing.
 (the marble = the marble; the marble rolling $\neq$ the marble standing)

Stage-level predicates having variables for stages

- (26) a. na-nun [ku kwusul-i kwulleka-nun] kes-ul cap-ass-ta.
na-nun the marble-*nom* roll-*adv* thing-*acc* catch-*pst-dec*
'I caught the ball which was rolling' (stage – stage)
- b. ?[ku kwusul-i kwulleka-nun] kes-i tantanha-ta.
the marble-*nom* roll-*adv* thing-*nom* hard-*dec*
'The ball which is rolling is hard' (stage – individual)
- (27) a. ?na-nun [ku kwusul-i tantanha-n3] kes-ul cap-ass-ta.
na-nun the marble-*nom* hard-*adv* thing-*acc* catch-*pst-dec*
'I caught the ball which was hard' (individual – stage)
- b. ?[ku kwusul-i tantanha-n3] kes-i pala-n3 sayk-i-ta.
the marble-*nom* hard-*adv* thing-*nom* blue-*adv* color-be-*dec*
'The ball which is hard is blue' (individual – individual)
- A relative clause must denote a property of entities which a variable range over.
 - The IHs are definite and there is no other place for a variable which ranges over things.
 - A difference between a stage-level predicate and an individual-level predicate

is that the former has an extra spatio-temporal argument, assuming Kratzer (1995).

- But an IHRC with a definite IH does not denote a property of the extra spatio-temporal argument, but of the IH.
- A spatio-temporal part of an IH is a stage involved in the event of the IHRC and the embedding clause.
- An IHRC can denote a property of stages when it has a stage-level predicate.
- This explains why an IHRC must have a stage-level predicate.
- The predicate in the matrix clause must also be a stage-level predicate because the NP modified by the IHRC is a stage.

Cf.

- (28) ku [e kwulleka-n3] kwusul-i tantanha-ta.
the roll-*adv* marble-*nom* hard-*dec*
'The ball which is rolling is hard.'

(In an EHRC the variable ranges over individuals.)

Type-mismatch and Type-shifting

- (29) mina-ka nemeci-ess-ta.
Mina-*nom* fall-*pst-dec*
'Mina fell.'

Here *fell* requires a stage of an individual, but *Mina* denotes an individual.

Type shifting rule:

- (30) If a stage-level predicate P applies to an individual d, then
 $P(d) = \exists x1[R(x1, d) \wedge P(x1)]$ ('R(a,b)' means 'a is a stage of b'.)

Thus the meaning of (29) can be expressed like the following:

- (31) $fell(Mina) = \exists x1[R(x1, Mina) \wedge fell(x1)]$

Question: How is the variable x1, once closed by an existential quantifier, opened again to give the meaning of a property?

Answer: **Existential disclosure**

Existential disclosure

A relativizer (*u*)*n* is a complementizer which takes a proposition and yields a property of entities.

Thus it has the meaning of **existential disclosure** proposed by Dekker (1993).²

- (32) $[(u)n_{C_{rel}}]^{w,g} = \lambda p \lambda x[p \wedge y=x]$

(p is a variable for propositions and y is **existentially bound** in p.)

Application of this rule to the IHRC in (22):

- (33) $[[mina-ka-ka-n_{imp}-\theta]_{T_1-\theta}]^{w,g}_{M_{rel}}]$
 $= \exists e \exists x1[R(x1, m) \wedge t_e \subseteq \tau(e) \wedge go(x1)(e)(w^*)]$ (assuming $g(i) = t_i$)

²The equation $y=x$ must go into the scope of the existential quantifier. This is ensured when we assume dynamic semantics, just as Dekker (1993) did.

1. $\lambda x1[\text{Bell}^T t = \tau(\tau(e); \text{fall}(m)(e)) \wedge e = \tau x]$ (E: existential quantifier in dynamic semantics)
 $= \lambda x1[\text{Bell}^T(\tau(e); \tau(\text{fall}(m)(e)) \wedge (e=x)) \wedge \tau(\text{export})]$
 $= \lambda x3\lambda t1[(\tau(e); \tau(\text{fall}(m)(e)) \wedge (e=x)) \wedge \tau(\text{import})] \wedge \tau(\text{export})$ (E: existential quantifier in predicate logic)
 $= \lambda x3\lambda t1[(\tau(e); \tau(e) \wedge \tau(\text{fall}(m)(e) \wedge \tau(e=x)) \wedge \tau(\text{import})]$
 $= \lambda x1\tau(e) \wedge \text{fall}(m)(e) \wedge e=x \wedge \tau(\text{import})]$

Properties of events or stages or facts

- (34) a. $\llbracket \text{mina-ka ka-n}_{\text{imp}}^{-\theta_{T-}\theta_{M_w}}\text{-un}_{C_{\text{rel}}} \rrbracket^{w*,g}$
 $= \lambda p \lambda x [p \wedge x1=x](\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)])$
 $= \lambda x [\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)] \wedge x1=x]$
 $= \lambda x1 \exists e [R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)] \text{ (a property of stages)}$
 b. $\llbracket \text{mina-ka ka-n}_{\text{imp}}^{-\theta_{T-}\theta_{M_w}}\text{-un}_{C_{\text{rel}}} \rrbracket^{w*,g}$
 $= \lambda p \lambda x [p \wedge e=x](\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)])$
 $= \lambda x [\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)] \wedge e=x]$
 $= \lambda e \exists x1 [R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w^*)] \text{ (a property of events)}$
 c. $\llbracket \text{mina-ka ka-n}_{\text{imp}}^{-\theta_{T-}\theta_{M_w}}\text{-un}_{C_{\text{rel}}} \rrbracket^{w*,g}$
 $= \lambda p \lambda x [p \wedge w=x](\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w)])$
 $= \lambda x [\exists e \exists x1[R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w)] \wedge w=x]$
 $= \lambda w \exists e [R(x1,m) \wedge t_r \subseteq \tau(e) \wedge \text{go}(x1)(e)(w)] \text{ (a property of facts)}$

(35) $\llbracket e_x \text{ ka-nu} \rrbracket^{w*,g} \text{ (x is free) (EHRC)}$

$$= \exists e \exists t [t \subseteq \tau(e) \wedge R(x1, g(x)) \wedge \text{go}(x1)(e)(w)]$$

$$\llbracket e_x \text{ ka-nu-n} \rrbracket^{w*,g} \text{ (x is bound by } (u)n \text{)}$$

$$= \lambda y [\exists e \exists t [t \subseteq \tau(e) \wedge R(x1, x) \wedge \text{go}(x1)(e)(w)] \wedge x=y]$$

$$= \lambda y [\exists e \exists t [t \subseteq \tau(e) \wedge R(x1, y) \wedge \text{go}(x1)(e)(w)]]$$

Properties of IHRCs of Type I

- The predicates in the IHRC and the embedding clause are stage-level predicates, including state verbs.
 - The IH can be definite.
 - The adnominal ending is mostly *nun* or *(u)n3*.
- (36) $?na\text{-nun}$ [ku kwusul-i kwulleka-n2] kes-ul cap-ass-ta.
 na-nun the marble-*nom* roll-*adv* thing-*acc* catch-*pst-dec*
 'I caught the ball when it had rolled down.'
- If the adnominal ending is $n1_{pf}$, the IH is in the stage of a resulting state:
- (37) *inbo-nun* mina-ka tachi-n1 kes-ul pwuchwukhay-ass-ta.
inbo-top Mina-*nom* hurt-*adv* thing-*acc* hold-*pst-dec*
 'Inbo held Mina when she was hurt.'
- The IHRC restricts a stage of the IH involved in the situation in it.
 - The same stage of the IH is involved in the events of the IHRC and the embedding clause. (in (26a))
- (38) $\exists x1 \exists x2 [R(x1, \text{the ball}) \wedge \text{was-rolling}(x1) \wedge R(x2, \text{sp}) \wedge \text{caught}(x1)(x2)]$

3.2 An IHRC of Type II as denoting a property of individuals

An IHRC of Type II can take an individual-level predicate.

(In a context where someone explains how to avoid tears when you cut onions,)

- (39) [khal-i cal tu-nun] ke(s)-(u)lo calu-myen nwun-i hwaksilhi tel
 knife-*nom* well cut-*adv* thing-with cut-if eyes-*nom* surely less
 mayp-supni-ta.
 pungent-*hon-dec*
 'If you cut (onions) with a sharp knife, they are less pungent on your eyes.'

(In a context where someone explains why green tea tastes bitter)

- (40) [nokcha-ka coh-un3] kes-un ku ssu-n3 mas-i tel ha-pni-ta.
 green.tea-*nom* good-*adv* thing-top its bitter-*adv* taste-*nom* less do-*hon-dec*
 'Green tea which is good tastes less bitter.'

(In a context where someone looks for a reading desk good for him)

(After it is said "Do you have a strong reading desk which is adjustable?")

- (41) [chayk-i twukewu-n3] ke(s)-i manh-ase-yo.
 book-*nom* thick-*adv* thing-*nom* many-*dec-hon*
 'Thick books are many' (I have many thick books.)

(In a context where someone explains about a house)

- (42) kayincekulo [cip-i nac-un3] ke(s)-i ancengtoy poi-m.
 personally house-*nom* low-*adv* thing-*nom* stable look-*ger*
 'Personally a house which is low looks stable.'

(In a context where someone recommends a coffee)

- (43) [khephi-ka cinha-n3] ke(s-ul) cohaha-nun pwun-i-lamyen (ikes-i)
 coffee-*nom* strong-*adv* thing(-*acc*) like-*adv* person-be-if (this-*nom*)
 koaynchan-ultus.
 okay-likely
 'If you are a person who likes strong coffee, this will be good for you.'

- (44) a. $[_{NP} \text{ cip-i nac-un3 kes-i}] \text{ cin-ey kangha-ta.}$
 house-*nom* low-*adv* thing-*nom* earthquake-at strong-*dec*
 'A house which is not tall is resistant to earthquakes.'
- b. na-nun $[_{NP} \text{ khephi-ka hyang-i cinha-n3 kes-ul}] \text{ masi-ess-ta.}$
 I-top coffee-*nom* smell-*nom* strong-*adv* thing-*acc* drink-*pst-dec*
 'I drank coffee which had strong smell.'

Properties of IHRCs of Type II

- The IHRC can have an individual-level predicate.
- The IH must be indefinite.

- (45) a. $??$ [_{NP} **ku** cip-i nac-un3 kes-il] cicin-ey kangha-ta.
the house-*nom* low-*adn* thing-*nom* earthquake-at strong-*dec*
'The house which is not tall is resistant to earthquakes.'
b. $??$ [_{NP} **ku** kkun-i sinchwukseng-i iss-nun kes]-ulo chaik-ul
the rope-*nom* flexibility-*nom* have-*adn* thing-with book-*acc*
mwuk-ess-ta.
te-*pst-dec*
'I tied the books up with the rope which is flexible.'

- The IHRC restricts an individual which the IH ranges over.
 - The embedding clause can have an individual-level or stage-level predicate.
- (46) na-nun [kwusul-i tantanha-n3] kes-ul cap/senhohay-(a)ss-ta.
na-nun marble-*nom* hard-*adn* thing-*acc* catch/ prefer-*pst-dec*
'I caught/ preferred the/a ball which was hard.'

- No restriction on the adnominal ending.

- (16) [[chayk-i suywu-{n3/(u)}/ess-ul/ess-te-n}] kes-ul] kol-ala.
book-*nom* easy-*{impr/mod/pst.mod/pst-tr-rel}* thing-*acc* select-*imp*
'Choose books which {are, will be, must have been, was} easy.'

Interpretation of a IHRC of Type II

- (47) na-nun [_{NP} cip-i nac-un3 kes-ul] sa-ss-ta.
I-top house-*nom* low-*adn* thing-*acc* buy-*pst-dec*
'I bought a low house.'

- (48) $\llbracket_{CP} \text{cip-i nac-} \langle \lambda_{\text{apt}} \lambda_{\text{T}} \lambda_{\text{M}} \text{-un}_C \text{ kes-ul} \rangle \rrbracket^{w*,g} =$
 $\lambda x [\exists y \exists e [t_e \subseteq \tau(e) \wedge \text{house}(y) \wedge \text{low}(y)(e)(w^*)] \wedge y=x]$
 $(= \lambda x \exists e [t_e \subseteq \tau(e) \wedge \text{house}(x) \wedge \text{low}(x)(e)(w^*)])$

The IHRC denotes a property of being a house which is not tall.

3.3 IHRCs of Type III

There are examples in which the IHRCs have stage-level predicates while the embedding clauses have stage-level or individual-level predicates:

- (49) a. [(ku) sakwa-ka ssek-un1] kes-i swul sengpwun-i iss-ess-ta.
the apple-*nom* rot-*adn* thing-*nom* wine ingredient-*nom* be-*pst-dec*
'The apple which rotted contained alcohol.'
b. [ku kamca-lul thwuyki-n1] kes-i tel cca-ss-ta.
the potato-*acc* fry-*adn* thing-*nom* less salty-*pst-dec*
'The potato that was fried was less salty.'

- The IH can be definite:
 - The adnominal ending is always $(u)n1$.
 - The predicates in the IHRCs are stage-level predicates which take affected arguments.
 - The affected arguments can be taken to be new entities, so we can think that there are some implicit arguments involved.
- (50) [(ku) sakwa-ka ssek]

$= \exists e \exists y \exists x1 [R(x1, \text{the apple}) \wedge \tau(e) \prec u \wedge \text{rotted}(x1)(e)(w) \wedge \text{BECOME}(\text{the apple}, y)]$

Cf. There are cases where $(u)n1$ is used and the affected argument is not a new entity:

- (37) inho-nun mina-ka tachi-n1 kes-ul pwuchwukhay-ss-ta.
Inho-top Mina-*nom* hurt-*adn* thing-*acc* hold-*pst-dec*
'Inho held Mina, when she was hurt.'

- When the IH is definite, there might be only one entity that has the property denoted by an IHRC of Type III, but it still is a property of individuals, not stages.

- IHRCs denote properties of resulting things.

- (51) a. [(ku) sakwa-ka ssek-un1]
 $= \lambda z \exists e \exists x1 [R(x1, \text{the apple}) \wedge \tau(e) \prec u \wedge \text{rotted}(x1)(e)(w) \wedge \text{BECOME}(\text{the apple}, y) \wedge y=z]$
 $= \lambda z \exists e \exists x1 [R(x1, \text{the apple}) \wedge \tau(e) \prec u \wedge \text{rotted}(x1)(e)(w) \wedge \text{BECOME}(\text{the apple}, z)]$

- There are cases where you can use common noun head nouns, if the resulting things have a separate name.

(52) inho-nun [mina-ka ku soykoki-lul pwul-ey mali-nl]
Inho-top Mina-nom the beef-acc fire-at dry-adv
kes/ywulkpho-ul mek-ess-ta.
thing/jerky-acc eat-psf-dec

Inho ate the thing/jerky that Mina dried the beef into over fire.'

(53) [swucungki-ka ungkyelha-n] mwul/kes-ul masi-ess-ta.
steam-nom condense-prf water/thing-acc drink-psf-dec

'(I) drank the water/stuff which steam was condensed into.

(54) $\lambda y \exists x, y, x1, e [\text{steam}(x)(w^*) \wedge \{ \exists x1 [\tau(e) < t, \wedge R(x1, x) \wedge \text{condense}(x1)(e)(w^*)] \}$
 $\wedge \text{BECOME}(x, y) \wedge \text{water}(y)(w^*)]$

In this respect, this can be taken to be an EHRC which involves an implicit argument.

Comparisons of the three types

	Type I	Type II	Type III
IH	a person/a thing/stuff	a thing/stuff	a thing/stuff
definiteness of IH	definite/indefinite	indefinite	definite/indefinite
embedded predicate	stage-level	stage-/individual-level	stage-level
embedding predicate	stage-level	stage-/individual-level	stage-/individual-level
meaning	a property of stages	a property of individuals	a property of individuals
adnominal ending	-nin /-(a)n/1/3	no restriction	-(a)n/1

Mixed cases

(55) a. inho-nun [sak-wa-ka ik-e teleci-nun] kes-ul mang-ulo
Inho-top apple-nom ripe-and fall-adv thing-acc net-in
pat-ass-ta. (Type I)
take-psf-dec

Inho took apples in a net when they ripened and fell.'

b. inho-nun [sak-wa-ka ik-e teleci-nun] kes-man phal-ass-ta.
Inho-top apple-nom ripe-and fall-adv thing-acc sell-psf-dec
(Type II)

'Inho sold apples which ripened and fell.'

(56) [sakwa-ka ik-e teleci-nun]

a. $\lambda y \exists x \exists x1 \exists e [\text{apple}(x)(w) \wedge R(x1, x) \wedge \text{ripe}(x1)(e)(w) \wedge \text{fall}(x1)(e)(w) \wedge x1 = y]$

b. $\lambda y \exists x \exists x1 \exists e [\text{apple}(x)(w) \wedge R(x1, x) \wedge \text{ripe}(x1)(e)(w) \wedge \text{fall}(x1)(e)(w) \wedge x = y]$

(57) a. [kam-i cal ik-unl] kes-ul mali-ki siakhay-ss-ta.
persimmon-nom well ripe-adv thing-acc dry-nml begin-psf-dec
'I began to dry persimmons when they ripened well.'

b. [kam-i cal ik-unl] kes-ul kola-ss-ta.
persimmon-nom well ripe-adv thing-acc choose-psf-dec
'I chose persimmons which ripened well.'

c. [kam-i cal ik-unl] kes/yensi-(n)un kenkang-ey coh-ta.
persimmon-nom well ripe-adv thing/yensi-top health-for good-dec
'Persimmons that have ripened well are good for health.'

(58) a. $\lambda y \exists x \exists x1 \exists e [\text{persimmon}(x)(w) \wedge R(x1, x) \wedge \text{ripen}(x1)(e)(w) \wedge x1 = y]$

b. $\lambda y \exists x \exists x1 \exists e [\text{persimmon}(x)(w) \wedge R(x1, x) \wedge \text{ripen}(x1)(e)(w) \wedge x = y]$

c. $\lambda z \exists x \exists y \exists x1 \exists e [\text{persim}(x)(w) \wedge R(x1, x) \wedge \text{ripen}(x1)(e)(w) \wedge \text{BECOME}(x, y) \wedge y = z]$

3.4 IHRCs in propositional attitude contexts

Type I

- (59) $?inho-nun$ [[mina-ka nemeci-n-ta-ko sayngkaktoy-nun]_{CP} kes-ul]_{NP}
 Inho-top Mina-nom fall-*impr-dec-cmp* be.thought-*adn* thing-*acc*
 cap-ass-ta. (IHRC)
 hold-*pst-dec*
 ‘Inho held Mina when he thought she was falling down.’

The IH needs to have wide scope so a stage of Mina can be involved in the event of the embedding clause. But stages have narrow scope:

- (60) [mina-ka nemeci-n-ta-ko sayngkaktoy-nu]_{CP}^{w,g}
 $= \exists e[think_{w,g}(\exists e[fall(m)(e)(w)])(sp)(e)(w^*)]$
 $= \exists e[think_{w,g}(\exists e\exists x[[R(x)l.m] \wedge fall(x)l](e)(w)])(sp)(e)(w^*)]$ (type-shifting)

(Here I assume that *think* is a universal quantifier over epistemic alternatives *w* of *w** to *sp*.)

Technically x1 could be opened by existential disclosure, but it cannot be the argument of the verb in the embedding clause, because x1 may not exist in the actual world *w**.

3. Three types of IHRCs

Three types of IHRCs in Korean and their interpretations

Type II

- (61) a. na-nun [[cip-i nac-ta-ko] sayngkaktoy-nun] kes-ul
 I-top house-nom low-*dec-cmp* be.thought-*adn* thing-*acc*
 sa-ss-ta.
 buy-*pst-dec*
 ‘I bought a house which I thought was not tall.’
 b. [[cip-i nac-ta-ko sayngkaktay-n- θ_{T_i} - θ_{N_i} -nun]
 i. $\lambda y[\exists e\exists x[house(x)(w^*) \wedge t_i \subseteq \tau(e) \wedge think_{w,g}(low(x)(w)](sp)(e)(w^*) \wedge x=y]$
 ii. $?? \lambda y[\exists e[t_i \subseteq \tau(e) \wedge think_{w,g}(\exists x[house(x)(w) \wedge low(x)(w)])(sp)(e)(w^*) \wedge x=y]$

Type III

- (62) $?inho-nun$ [[ku swucungki-ka ungkyelhay-ss-ta-ko sayngkaktay-nun]
 Inho-top the steam-nom condense-*prf-dec-cmp* think-*adn*
 kes-ul masi-ess-ta.
 thing-*acc* drink-*pst-dec*
 ‘Inho drank what he thought the steam was condensed into.’

The resulting entities are implicit arguments, which cannot have wide scope, as pointed out by Fodor & Fodor (1980) and Condoravdi & Gawron (1996).

4. Meaning of kes

Three types of IHRCs in Korean and their interpretations

4 Meaning of kes

Previous analyses:

In the E-type anaphora analysis of IHRCs proposed by Hoshi (1995), Shimoyama (1999, 2001), M.J. Kim (2007), or the analysis of Grosu (2010), a NP modified by an IHRC is assumed to be **always definite**.

Problem 1:

A NP modified by an IHRC is bound by quantificational adverbs or the generic operator.

- (63) inho-nun nunl mina-ka nemeci-nun kes-ul cap a.cwu-n-ta.
 Inho-top always Mina-nom fall-*adn* thing-*acc* hold-*impr-dec*
 ‘Inho always holds Mina whenever she falls.’
ALWAYS_{x1,e1,t} [t_C $\subseteq \tau(e1) \wedge R(x1,m) \wedge fall(x1)(e1)(w^*)$
 $\rightarrow \exists x2,e2[R(x2,c) \wedge hold(x1)(x2)(e2)(w^*)]]$

- (64) inho-nun pothong mina-ka nuccamea-nun kes-ul an kkeywu-n-ta.
 Inho-top usually Mina-nom oversleep-*impr* thing-*acc* not wake-*impr-dec*
 ‘Inho usually does not wake Mina when she oversleeps.’
GEN_{w,x1,x2,e1} [[R(x1,m) $\wedge$ oversleep(x1)(e1)(w) $\wedge$ R(x2,i) $\wedge$ C(x2,e1)]
 $\rightarrow \neg \exists e2[wake(x1)(x2)(e2)(w)]]$

C(x2,e1): C is an appropriate condition on situations in which a stage of Inho is normally in the event e1.

4. Meaning of kes

Three types of IHRCs in Korean and their interpretations

Problem 2: can be definite or indefinite

When there is no operator like the generic operator, an existential reading obtains.

- (65) a. inho-nun mina-ka nuccamea-nun kes-ul kkeywu-ess-ta.
 Inho-top Mina-nom oversleep-*impr* thing-*acc* wake-*pst-dec*
 ‘Inho woke Mina when she overslept.’
 b. $\exists x1,x2,e1,e2[t_i \subseteq \tau(e1) \wedge R(x2,m) \wedge oversleep(x2)(e1)(w^*) \wedge \tau(e2)=t_i, < t_u$
 $\wedge R(x1,c) \wedge wake(x2)(x1)(e2)(w^*)]$

Definite NPs

- (66) mina-ka nemeci-ko iss-ess-ta. [mina-ka nemeci-nun] kes-ul
 Mina-nom fall-*prg-pst-dec* Mina-nom fall-*adn* thing-*acc*
 cap-ass-ta.
 hold-*pst-dec*
 ‘Mina was falling. (I) held Mina when she was falling.’
 (67) cip-i khu-n3 kes-kwa cak-un3 kes-i iss-ess-ta. [cip-i
 house-nom big-*adn* thing-and small-*adn* thing-nom be-*pst-dec* house-nom
 cak-un3] kes-i te pissas-ta.
 small-*adn* thing-nom more expensive-*pst-dec*
 ‘There’s a big and a small house. The small house was more expensive.’

Indefinite NPs

- (68) a. na-nun [pang-i atamha-ko coyongha-n3] kes-ul
 I-top room-*nom* small-and quiet-*adv* thing-*acc*
chac-ass-ta.
 find/look-for-*pst-dec*
 'I found / {looked for} a small and quiet room.'
 b. kulena **kulehan** pang-un ups-ess-ta. (with *looked for*)
 but such room-top not-be-*pst-dec*
 'But there was no such room.'

The NP modified by the IHRC is a **non-specific** indefinite.

Double IHRCs of Type II

The IH of an IHRC of Type II must be indefinite.

- (69) [[**khemphyuthe-ka cak-un3 kes-il**] sengnung-*kkaci* coh-un3
 computer-*nom* small-*adv* thing-*nom* performance-even good-*adv*
 kes-ul] sa-ss-ta.
 thing-*acc* buy-*pst-dec*
 'I bought a computer which was small but even had a good performance.'

Here the IH, which is indefinite, also includes an IHRC.

Bare NPs in Korean

- (70) a. mina-nun **haksaying-tul**-ul manna-ss-ta.
 Mina-top student-pl-*acc* meet-*pst-dec*
 'Mina met some / the students.'
 b. **haksaying-tul**-un pothong cemsim-ul ilccik mek-nun-ta.
 student-pl-top generally lunch-*acc* early eat-*prf-dec*
 'Students generally eat lunch early.'

Assumptions:

- Basic denotations of bare NPs are properties.³
- Used as arguments of predicates, they do not fit with the type required by the predicates. (ParTEE (1987), Chierchia (1998), Krifka (2004), etc):

$$(71) \quad \begin{array}{lcl} \text{I-operation : } <e,t> \rightarrow e: & & P \rightarrow \iota x[P(x)] \\ \text{E-operation : } <e,t> \rightarrow <<e,t>,t>: & & P \rightarrow \lambda Q\exists x[P(x) \wedge Q(x)] \end{array}$$

A phrase headed by *kes* is just like a bare NP.

³Chierchia (1998) assumes that in languages like Chinese a noun denotes a kind, which can be freely converted into a property. Korean has a lot in common with Chinese. Thus we can assume that a property is derived from a kind.

Semantics of *kes*

kes is just a nominalizer which just changes the syntactic category, with no semantic contribution, except for the presupposition that the property applies to non-humans:

$$(72) \quad [\text{kes}] = \lambda P . P$$

This is like a null noun in English.

- (73) a. A: Three pints of bitter $\emptyset$.
 B: Best $\emptyset$ or ordinary $\emptyset$?
 b. We've just bought a new car. It's an automatic $\emptyset$.

$\emptyset$ = beer, car, ... (determined contextually)

- (74) A: enu cha sal-kka? B: kkamah-n3 **ke(s)**-lo ha-ca.
 which car buy-int black-*adv* thing-to do-*lor*
 'A: Which car shall we buy? B: Let's buy the black (**one**).'

Alternative: $[\text{kes}] = \lambda x . \text{thing}(x)$

- the property of being a thing is not asserted.
- it always requires a modifier.

5 Distributions of three types of IHRCs

Two factors that determine the distributions of the three types of IHRCs:

- The defective noun *kes* is used to refer to some stuff, things or events, but not people.
- Subjects must denote individuals, while objects can denote either individuals or stages.

Assumption: The distinction of people and things is valid only when the denotations are individuals. Stages are things.

Implications on the distributions of IHRCs

- The IHs of IHRCs in subject positions cannot denote people.
- Since subjects cannot be stages, no IHRCs of Type I occur in subject positions, while IHRCs of Types II and III can.
- No restrictions on IHRCs in object positions.
- All three types of IHRCs can occur in object positions.
- If the IHs of IHRCs in object positions denote people, the predicates must be stage-level and the NPs headed by *kes* must denote their stages.

5.1 IHRCs in subject positions

No IHs denote persons in subject positions

Since *kes* denotes only things, not people, and subjects cannot be stages, it is predicted that the IH of an IHRC in a subject position must not be a person:

- (75) [pak-ey {cha-ka} / ??{haksayng-i}] s(u)-e-iss-nun] ke(s)-i ney
outside-*at* {car-*nom*} / {student-*nom*} stand-*prg-adn* thing-*nom* your
{cha/haksayng}-i-ta.
car / student-be-*dec*

‘The car / student standing outside is your car / student.’

- (76) [pak-ey {cha-ka} / ??{haksayng-i}] s(u)-e-iss-nun] ke(s)-i isanghan
outside-*at* {car-*nom*} / {student-*nom*} stand-*prg-adn* thing-*nom* strange
sol-i-lul nay-ss-ta.
sound-*acc* make-*pst-dec*

‘The car / student outside made a strange sound.’

IHRCs of Type II, but not of Type I, in subject positions

- (77) [cha-ka s-e-iss- θ_{I_1} - θ_{M_n} - $\Pi_{C_{rel}}$]
two readings:
 $\lambda y[\exists x, x1.e]_{\tau} \subseteq \tau(e) \wedge \text{car}(x) \wedge R(x1, x) \wedge \text{stand}(x1)(e)(w^*) \wedge z=y]$,
where (i) $z = x$, or (ii) $z = x1$

- i. $z=x$: the IHRC denotes a property of individuals, (allowed, as shown in (75))
ii. $z=x1$: the IHRC denotes a property of stages, (not allowed, as shown below)

- (78) a. ??[pak-ey ku cha-ka s(u)-e-iss-nun] kes-i nay cha-i-ta.
outside-*at* the car-*nom* stand-*prg-adn* thing-*nom* my car-be-*dec*
‘The car standing outside is my car.’
b. ??[pak-ey ku cha-ka s(u)-e-iss-nun] kes-i isanghan sol-i-lul
outside-*at* the car-*nom* stand-*prg-adn* thing-*nom* strange sound-*acc*
nay-ss-ta.
make-*pst-dec*

‘The car standing outside made a strange sound.’

The IHRCs have stage-level predicates and the IHs are definite. So the IHRCs should denote properties of stages, which is not allowed in the subject position.

IHRCs of Type III in subject positions

It is possible that the predicates are stage-level and the IHs are definite.

- (79) a. [ku yulchang-i [kayci-n1] kes-i patak-ey teleci-ess-ta.
the window-*nom* break-*adn* thing-*nom* floor-on fall-*pst-dec*
‘The thing(s)(pieces) which the window broke into fell on the floor.’
b. [ku sakwa-lul seek-hi-(u)n1] kes-i akchwuy-lul
the apple-*acc* rot-make-*adn* thing-*nom* stench-*acc*
phwungki-ess-ta.
give.off-*pst-dec*

‘The apple which I made rotten gave off a bad smell.’

- c. [[ku swucungki-ka ungkyelha-n1] kes / mwul-il] mal-k-ass-ta.
the steam-*nom* condense-*adn* thing / water-*nom* clear-*pst-dec*
‘The thing / water that the steam was condensed into was clear.’

5.2 IHRCs in object positions

In an IHRC of Type I, the IH can denote a person.

In this case, the IHRC will denote a property of stages:

- (80) inho-nun [mina-ka namca-lul manna-nun] kes-ul
Inho-*top* Mina-*nom* man-*acc* meet-*adn* thing-*acc*
teyilkonaao-ass-ta.
take.outside-*pst-dec*
‘Inho took Mina outside, when she was meeting a man.’
‘Inho took a man outside, when Mina was meeting him.’

take out is a stage-level predicate.

- (81) ?*inho-nun [mina-ka namca-lul manna-nun] kes-ul mit-ess-ta.
Inho-*top* Mina-*nom* man-*acc* meet-*adn* thing-*acc* believe-*pst-dec*
‘Inho believed the / a man who Mina was meeting.’

The embedding predicate *mit* ‘believe’ requires a human individual as its object. However, the sentence is odd. ((81) cannot be Type II because of *kes*, nor Type I because of *mit*.)

IHRCs of Type II in object positions

Even in the object position, if the IHRC is Type II, the IH cannot be human.

– the IH is human

- (82) ?*in_{ho}-nun [namca-ka khi-ka khu-n3] kes-ul manna-ss-ta.
In_{ho}-*top* man-*nom* height-*nom* tall-*adn* thing-*acc* meet-*pst-dec*
‘In_{ho} met a man who was tall.’

– the IH is non-human

- (83) na-nun [[khephi-ka cinha-n3] kes-ul] sa-n/ss-ta.
I-*top* coffee-*nom* strong-*adn* thing-*acc* buy-*impr*/ *pst-dec*
‘I buy/bought coffee which is/was strong.’

IHRCs of Type III in object positions

(79c) is an example in which an IHRC of Type III occurs in the object position and the predicate in the embedding clause denotes a long-lasting property which do not require a stage of an object.

- (84) ku kamca-lul thwuyki-n1 kes-ul mek-ess-ta.
the potato-*acc* fry-*adn* thing-*acc* eat-*pst-dec*
‘(I) ate the potato that was fried.’

6 No necessity of the relevancy condition

Kuroda’s (1976-77, 1992) Relevancy Condition and Fujii’s (1998) Simultaneity Condition (or, the Precedence Adjacency Condition)

Relevancy condition for Type I

- (85) [ku cha-ka kuttay eyncin-i kkeci-e-iss-nun] kes-ul sitong-ul
the car-*nom* then engine-*nom* stop-*impr-adn* thing-*acc* starter-*acc*
start-*pst-dec*
‘(I) started the engine of the car when it was turned off then.’
- (86) ?[ku cha-ka cikum eyncin-i kkeci-e-iss-nun] kes-ul nayil
the car-*nom* now engine-*nom* stop-*impr-adn* thing-*acc* tomorrow
sitong-ul kel-1.kes-i-ta.
starter-*acc* start-*mod-dec*
‘(I) will start the engine of the car tomorrow when it is turned off now.’

- The IH is definite.
- The predicate in the IHRC does not take an affected argument. $\Rightarrow$ Type I
- The relevancy condition must be satisfied.

No relevancy condition for Types II and III**Type II**

- (87) (*ku) kkun-i sinchwukseng-i-iss-nun kes-un il nyen hwu-ey-to
the rope-*nom* elasticity-*nom* have-*adn* thing-*top* one year later-at-also
an kkunec-1.kest-ta.
not be-cut-*mod-dec*
‘The/A rope which is elastic will not be broken even in a year.’
- (88) (*ku) cha-ka eyncin-i kkeci-e-iss-nun kes-i nayil
the car-*nom* engine-*nom* stop-*impr-adn* thing-*nom* tomorrow
chwulcenha-1 cha-i-ta.
participate-*adn.mod* car-be-*dec*
‘The car whose engine is turned off is going to participate in the race tomorrow.’

- The IH cannot be definite. $\Rightarrow$ Type II
- The IHRCs denote the properties of individuals, whether the predicates are stage-level or individual-level.
- The relevancy condition is not satisfied.

Type III

- (89) in_{ho}-nun [[mina-ka cikum ku kamca-lul thwuyki-nun]_{CP} kes-ul]_{NP}
In_{ho}-*top* Mina-*nom* now the potato-*acc* fry-*adn* nayil thing-*acc*
mek-ul.kesi-ta.
tomorrow eat-*mod-dec*
‘In_{ho} will eat tomorrow the potato which Mina is frying now.’
- The IH is definite.
 - The predicate has an affected argument which can be taken to be a new entity. $\Rightarrow$ Type III
 - The relevancy condition is not satisfied.
 - The IHRC denotes a property of individuals.
- (90) [[mina-ka cikum ku kamca-lul thwuyki-nun]_{CP}]_{NP}^{10*,9}
 $\lambda z \exists e, x_1, x_2, y [R(x_1, m) \wedge R(x_2, \text{the potato}) \wedge \text{fry}(e, x_1, x_2) \wedge \text{now} \subseteq \tau(e) \wedge$
BECOME(the potato, y)] $\wedge y = z]$
 $= \lambda y \exists e, x_1, x_2 [R(x_1, m) \wedge R(x_2, \text{the potato}) \wedge \text{fry}(e, x_1, x_2) \wedge \text{now} \subseteq \tau(e) \wedge$
BECOME(the potato, y)]

Here y ranges over new entities resulting from an affected argument.

Apparent exceptions

Use of past time expressions

- (91) inho-nun latto-ka **olay** **cen-ey** kocangna-n] kes-ul
 Inho-*top* radio-*nom* long time before-at break-*adn* thing-*acc*
 kochi-ko iss-ta.
 fix-*prg-dec*

‘Inho is fixing the radio which broke down long time ago.’

In IHRCs only accomplishment and achievement verbs come with (u)n1. (Y.B. Kim (2002)) They give rise to resulting states at the reference time. The states make them satisfy the relevancy condition. Activity verbs come with -(u)n2, state verbs with *ess-ten*.

- (92) ??inho-nun mina-ka il-ul ha-n2 kes-ul tow-ass-ta. (activity)
 Inho-*top* Mina-*nom* work-*acc* do-*adn* thing-*acc* help-*pst-dec*
 ‘Inho helped Mina when she had worked.’
- (93) ??inho-nun mina-ka chwuwu-ess-ten kes-ul an-ulo teyli-ko
 inho-*top* mina-*nom* cold-*pst-adn.tr* thing-*acc* inside-to take
 o-ass-ta. (state)
 come-*pst-dec*
 ‘Inho took Mina inside, when she was cold (outside).’

Present perfect with a past time

- (94) A: inho-nun eti-ey iss-nya?
 Inho-*top* where-at be-*int*
 ‘Where is Inho?’
- B: (ku-nun) ecey ttena-ss-e.
 (he-*top*) yesterday leave-*prf-dec*
 ‘He has left yesterday.’

In German, Dutch, Iceland or Italian perfect interpretations are compatible with past time expressions, as pointed out in Giorgi, A. and F. Pianesi (1998), McCawley (1971), Musan (2001), Portner (2003), etc.

(One example from German)

- (95) Hans ist {gestern um zehn/jetzt} weggegangen. (Musan 2001)
 Hans is {yesterday at 10/now} left
 ‘Hans has left yesterday at 10/now.’

Here *gestern um zehn* and *jetzt* can be assumed to restrict the event time and the reference time, respectively.

Creation verbs

- (96) inho-nun mina-ka **ecey** (??ku) phyenci-lul ssu-n1 kes-ul
 Inho-*top* Mina-*nom* yesterday the letter-*acc* write-*adn* thing-*acc*
 pwuch-ess-ta.
 post-*pst-dec*
 ‘Inho posted the letter Mina wrote yesterday.’

- (97) [[mina-ka **ecey** phyenci-lul ssu-n]]^{tr*,d}
 = $\lambda y[\exists x1.x[R(x1,m) \wedge \text{letter}(x) \wedge \text{write}(e,x1,x) \wedge \tau(e) \subseteq \text{yesterday}] \wedge x=y]$

- A creation verb involves a variable for individuals, not one for stages.
- The IHRC denotes a property of individuals.
- The relevancy condition is not satisfied.

(Creation verbs behave like predicates in IHRCs of Type III.)

The relevancy condition allows a lot of exceptions, but the exceptions are systematically explained in the analysis here.

Summary

- In Type I, when a spatio-temporal stage participates in the event of an IHRC and that of its embedding clause, the two events must be simultaneous or temporally adjacent.
- In Type II, the IHRCs denote properties of individuals, and individuals are not restricted spatio-temporally.
- Predicates in Type III and creation verbs bring new entities into being, in which case no particular stages are involved in the events of IHRCs.
- No separate relevancy condition is necessary, because
 - i. when it is necessary, it just follows from the fact that the same stage is involved in the situations of both the IHRC and the embedding clause and
 - ii. when it is not necessary, no stages are involved in IHRCs

7 Quantifiers as IHS in IHRCs

- (98) ?John-un [[taypwupwun-uy/motun haksaying-(tul)-i nemeci-n-un]
John-top [[most-of/every student-(pl)-nom fall-impv-rel]
kes]-ul cap-ass-ta.
thing]-acc hold-*pst-dec*.
'John held most/every students when they were falling.'

IHRCs with existentially quantified NPs are better:

- (99) John-un [[{myech/sey} myeng-uy haksaying-i nemeci-n-un] kes]-ul
John-top [[{some/three} unit-of student-nom fall-impv-rel] thing]-acc
cap-ass-ta.
hold-*pst-dec*
'John held some/three students when they were falling.'

In (98), the quantifier seems to have wide scope over the relative clause:

- (100) a. $MOST_x(\{x\} \text{ students}(x))(\{x\} \text{ John held } x \text{ when } x \text{ was falling})$
b. John held ($MOST_x(\{x\} \text{ student}(x))(\{x\} \text{ student}(s), \text{ falling}(x))$)

The second reading is very strange.

Strong quantifiers in IHRCs have narrow scope

The negative polarity item is not licensed across an IHRC.

- (101) *John-un [[anwu haksaying-to nemeci-n-un] kes]-ul cap-a-cu-cl
John-top [[any student-and fall-impv-rel] thing]-acc hold-give-*imnl*
anh-ass-ta.
not-*pst-dec*
'John didn't hold any students who were falling.'

A quantifier in an IHRC has narrow scope:

- (102) ?hwu salam-un [[taypwupwun-uy/motun haksaying-(tul)-i
two people-top [[most-of/every student-(pl)-nom
nemeci-n-un] kes]-ul cap-ass-ta.
fall-impv-rel] thing]-acc hold-*pst-dec*.
'Two people held most/every student(s) when they were falling.'
two > most, all
Nevertheless, (98) is not that bad.

A quantifier introduces another discourse referent

- (103) a. Every student arrived early. *They* brought their books with them.
b. Most students arrived early. *They* brought their books with them.

In (103a, b), the pronoun *they* refers to the students who arrived early.

- (104) a. $\exists X.e[\text{students}(X)(w^*) \wedge \text{arrived_early}(X)(e)(w^*) \wedge$
EVERY($\lambda y[\text{student}(y)(w^*)](\lambda x[x \leq X \wedge \text{ATOM}(x)])$)]
b. $\exists X.e[\text{students}(X)(w^*) \wedge \text{arrived_early}(X)(e)(w^*) \wedge$
MOST($\lambda y[\text{students}(y)(w^*) \wedge y \leq Y \wedge \text{ATOM}(y)])(\lambda x[x \leq X \wedge \text{ATOM}(x)]$)

the interpretation of the IHRC in (98) on the basis of existential disclosure:

- (105) $\lambda Y[\exists X[\text{student}(X)(w^*) \wedge \forall x \leq X \exists x1.e[t_x \subseteq \tau(e) \wedge R(x1, x) \wedge \text{cry}(x1)(e)(w^*)] \wedge$
ALL/MOST($\lambda x[\text{student}(z)(w^*)](\lambda x[x \leq X \wedge \text{ATOM}(x)])$)] $\wedge X=Y]$
= $\lambda Y[\text{student}(Y)(w^*) \wedge \forall x \leq Y \exists x1.e[t_x \subseteq \tau(e) \wedge R(x1, x) \wedge \text{cry}(x1)(e)(w^*)] \wedge$
ALL/MOST($\lambda x[\text{student}(z)(w^*)](\lambda x[x \leq Y \wedge \text{ATOM}(x)])$)

This way of interpretation is not easy and (98) is a little odd.

8 Conclusion

- We divided IHRCs into three types depending on how they are interpreted.
 - An IHRC of Type I denotes a property of stages.
 - An IHRC of Type II denotes a property of individuals.
 - An IHRC of Type III denotes a property of individuals, but can be analysed as EHRCs.
- They commonly need a process of existential disclosure to get the right interpretations.
- The distribution of the three types of IHRCs is determined by the distinctions of (in)definiteness of IHS and the syntactic positions in which IHRCs occur, along with the meaning of *kes*.
 - NPs headed by *kes* in subject positions denotes individuals.
 - NPs headed by *kes* in object positions denotes individuals or stages.
 - Stages of individuals are not people.
 - NPs headed by *kes* do not denote people.
- Strong quantifiers introduce a hidden existential quantifier over individuals, the intersection of the restrictor and the nuclear scope.

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